

EDUCATION

•PhD in Physics : Complex Systems & Nonlinear Dynamics

Indian Institute of Science Education and Research (IISER), Thiruvananthapuram

2023 - 2028 (Expected)

- My current research focuses on adaptive dynamical networks, which serve as paradigmatic models for complex systems such as neuronal and social networks.
- I perform mathematical modeling, develop numerical simulations (Python and FORTRAN), and conduct mathematical analysis to investigate emergent dynamics in these systems.

•Integrated M.Sc. in Physics

National Institute of Technology (NIT), Patna

2017 - 2022

CGPA: 8.73/10

EXPERIENCE

•Teaching Assistant

IISER, Thiruvananthapuram

Aug 2023 - Present

- Classical Mechanics (Aug 2024–Dec 2024, Aug 2025–Present)
- Computational Techniques & Programming Languages (Jan 2025–May 2025)
- Numerical Methods (Jan 2024–May 2024)
- Optics (Aug 2023–Dec 2023)

•Junior Research Fellow

IISER, Thiruvananthapuram

Jul 2022 - Jan 2023

- Conducted research in nonlinear dynamics focusing on swarmlator dynamics.
- Developed and implemented numerical models and simulations using Python.

•Research Intern

Physical Research Laboratory, Department of Space, Government of India

Jan 2022 - Jun 2022

- Final Semester Project Training (FSTP-2022): Modeled and simulated surface charge accumulation on satellites due to plasma interactions. Developed a proof-of-concept for charge estimation, validated with existing data, and explored mitigation strategies considering plasma environment, geometry, and material properties. Outcomes aid satellite design and operation.
- The research project report: [Deleterious satellite charging and possible mitigation schemes](#)

•Freelance - Modeling and Simulation

Upwork (Remote)

Nov 2021 - Present

- Delivered computational modeling projects for physics and engineering domains.
- Developed Python-based simulations and visualizations for client needs.

•Subject Matter Expert (Physics)

Chegg Inc. (Freelance)

Mar 2019 - 2021

- Provided online tutoring and problem-solving in physics for global students.

PUBLICATIONS [[ALL PUBLICATIONS](#)]

•Heterogeneous nucleation due to adaptation disparity and phase lag in finite-size adaptive dynamical networks

Physical Review E

Apr 2025

•Disparity-driven heterogeneous nucleation in finite-size adaptive networks

Physical Review E

May 2024

TECHNICAL SKILLS

Languages: Python, FORTRAN

Libraries/Tools: NumPy, SciPy, Matplotlib, NetworkX, LaTeX

Areas: Complex Systems, Nonlinear Dynamics, Network Science, Computational Physics

Soft Skills: Scientific Writing, Teaching, Communication, Content Development, Problem Solving